

☒ EASY

☐ ALGEBRA

Algebra

10 questions · ~15 min

15

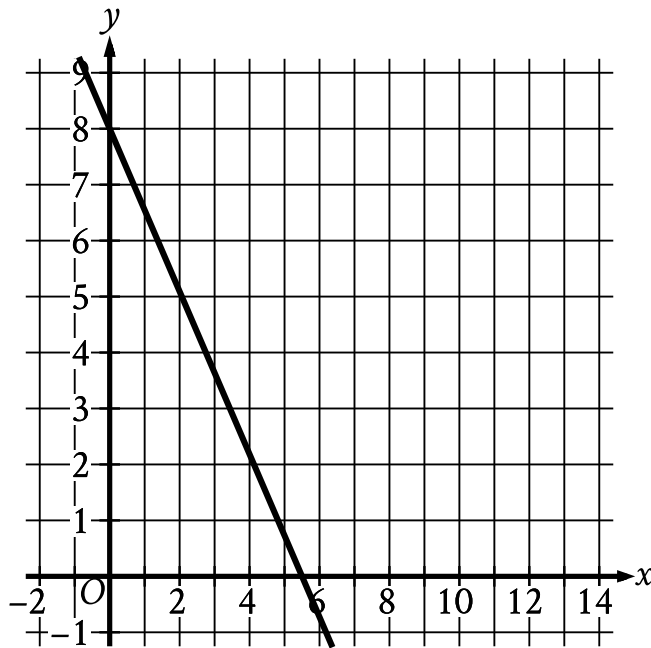
Q1

ALGEBRA

$$10 = 2x + 4$$

How many solutions exist to the equation shown above?

- A. None
- B. Exactly 1
- C. Exactly 3
- D. Infinitely many



- The line slants sharply down from left to right.
- The line passes through the following points:
 - (0, 8)
 - $(\frac{11}{2}, 0)$

The graph of the linear function f is shown, where $y = f(x)$. What is the y -intercept of the graph of f ?

- A. 0,0
- B. 0, $-\frac{16}{11}$
- C. 0, -8
- D. 0,8

$$\begin{aligned}y &= 2x + 3 \\ x &= 1\end{aligned}$$

What is the solution (x,y) to the given system of equations?

- A. $(1,2)$
- B. $(1,5)$
- C. $(2,3)$
- D. $(2,7)$

Davio bought some potatoes and celery. The potatoes cost \$ 0.69 per pound, and the celery cost \$ 0.99 per pound. If Davio spent \$ 5.34 in total and bought twice as many pounds of celery as pounds of potatoes, how many pounds of celery did Davio buy?

- A. 2
- B. 2.5
- C. 2.67
- D. 4

If $\frac{2n}{5} = 10$, what is the value of $2n - 1$?

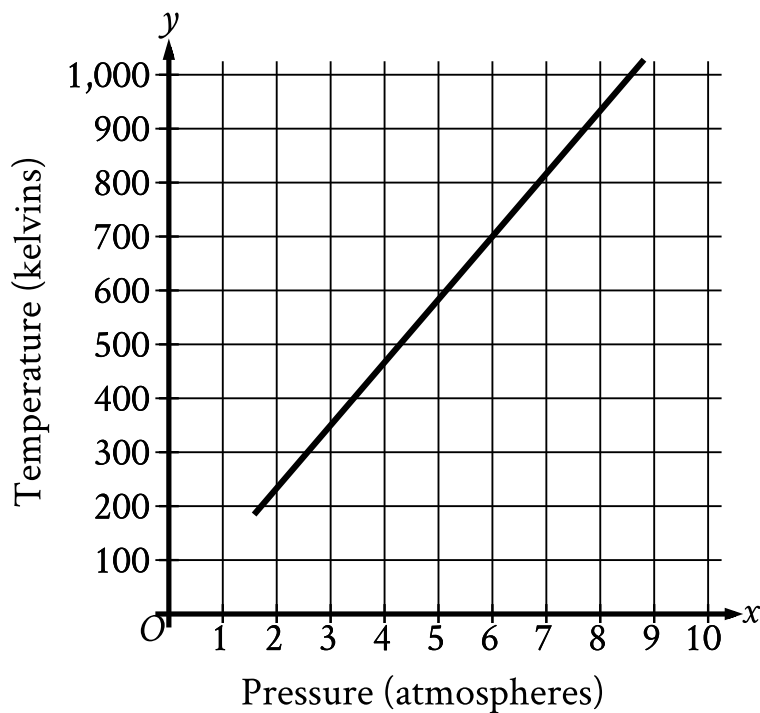
A. 24

B. 49

C. 50

D. 99

Oxygen gas is placed inside a tank with a constant volume. The graph shows the estimated temperature y , in kelvins, of the oxygen gas when its pressure is x atmospheres.



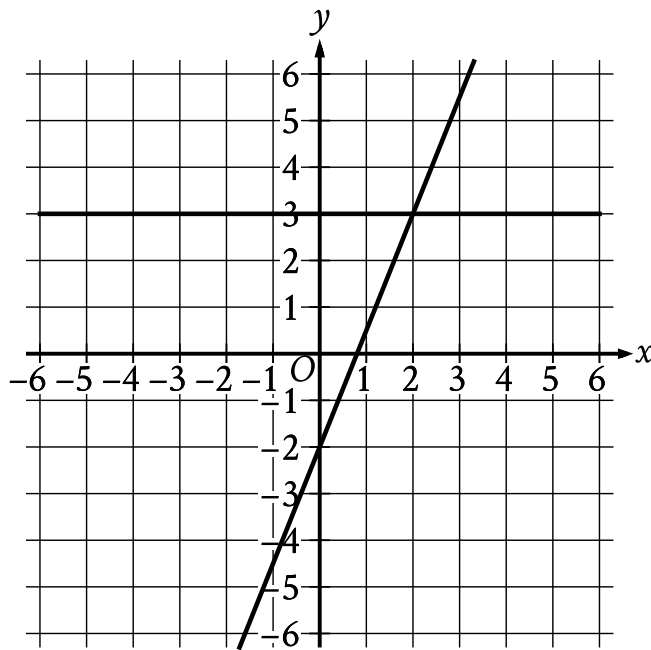
- The line slants sharply up from left to right.
- The line begins at the approximate point (1.6 comma 190).
- The line passes through the following approximate points:
 - (3 comma 350)
 - (6 comma 700)
 - (8 comma 933)

What is the estimated temperature, in kelvins, of the oxygen gas when its pressure is 6 atmospheres?

- A. 6
- B. 60

C. 700

D. 760



- For the first line in the system:
 - The line is horizontal.
 - The line passes through the following points:
 - (negative 4 comma 3)
 - (0 comma 3)
 - (2 comma 3)
- For the second line in the system:
 - The line slants sharply up from left to right.
 - The line passes through the following points:
 - (negative 1 comma negative 4.5)
 - (0 comma negative 2)
 - (2 comma 3)

The graph of a system of linear equations is shown. What is the solution x, y to the system?

A. 0, 3

B. 1, 3

C. 2, 3

D. 3, 3

A line in the xy -plane has a slope of $\frac{1}{9}$ and passes through the point $(0, 14)$. Which equation represents this line?

A. $y = -\frac{1}{9}x - 14$

B. $y = -\frac{1}{9}x + 14$

C. $y = \frac{1}{9}x - 14$

D. $y = \frac{1}{9}x + 14$

What value of p satisfies the equation $2p + 275 = 325$?

- A. 5
- B. 25
- C. 48
- D. 300

For the function f , the graph of $y = f(x)$ in the xy -plane has a slope of 3 and passes through the point $(0, -8)$. Which equation defines f ?

A. $f(x) = 3x$

B. $f(x) = 3x - 8$

C. $f(x) = 3x + 5$

D. $f(x) = 3x + 11$